

Examination Scheme for Semester-V

CA – Continuous Assessment

EoSE – End of Semester Examination

Regular Students –

Type of Examination	Course Code and Nomenclature	Scheme of Examination	Duration of Examination	Maximum Marks	Minimum Marks
Theory	UG0804 -BTH-75T-551 ANIMAL BIOTECHNOLOGY	CA	01 Hr	20 Marks	08 Marks
		EoSE	03 Hrs	80 Marks	32 Marks
Practical	UG0804 -BTH-75P-552 ANIMAL BIOTECHNOLOGY- PRACTICAL	CA	1 Hr	10 Marks	04 Marks
		EoSE	04 Hrs	40 Marks	16 Marks
Theory	UG0804 -BTH-75T-553 PLANT BIOTECHNOLOGY	CA	01 Hr	20 Marks	08 Marks
		EoSE	03 Hrs	80 Marks	32 Marks
Practical	UG0804 -BTH-75P-554 PLANT BIOTECHNOLOGY - PRACTICAL	CA	1 Hr	10 Marks	04 Marks
		EoSE	04 Hrs	40 Marks	16 Marks
Theory	UG0804 -BTH-75T-555 BIORESOURCE WASTE MANAGEMENT	CA	01 Hr	20 Marks	08 Marks
		EoSE	03 Hrs	80 Marks	32 Marks
Practical	UG0804 -BTH-75P-556 BIORESOURCE WASTE MANAGEMENT-PRACTICAL	CA	1 Hr	10 Marks	04 Marks
		EoSE	04 Hrs	40 Marks	16 Marks

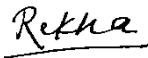
The theory question paper will consist of **two** parts, **A & B**.

PART-A: 20 Marks

Part A will be compulsory, having 10 very short answer-type questions (with a limit of 20 words) of two marks each.

PART-B: 60 Marks

Part B of the question paper shall be divided into four units comprising question numbers 2-5. There will be one question from each unit with an internal choice. Each question will carry 15 marks.

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UG0804 – Three/Four Year Bachelor of Science (Bio-Technology)									
				Bio-Technology 3 rd YEAR SEM-V		Credits			
#	L e v e l	S e m e s t e r	Type	Title		L	T	P	Total
1.	7	V	MJR	UG0804 -BTH-75T-551 ANIMAL BIOTECHNOLOGY		4	0	0	4
2.	7	V	MJR	UG0804 -BTH-75P-552 ANIMAL BIOTECHNOLOGY-PRACTICAL		0	0	2	2
3.	7	V	MJR	UG0804 -BTH-75T-553 PLANT BIOTECHNOLOGY		4	0	0	4
4.	7	V	MJR	UG0804 -BTH-75P-554 PLANT BIOTECHNOLOGY -PRACTICAL		0	0	2	2
5.	7	V	MJR	UG0804 -BTH-75T-555 BIORESOURCE WASTE MANAGEMENT		4	0	0	4
6.	7	V	MJR	UG0804 -BTH-75P-556 BIORESOURCE WASTE MANAGEMENT-PRACTICAL		0	0	2	2

BTH-75T-551 ANIMAL BIOTECHNOLOGY

Semester	Code of the Course	Title of the Course/Paper			NHEQ F Level	Credits
V	BTH-75T-551	ANIMAL BIOTECHNOLOGY			7	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				

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Prerequisites	Introductory course of Biotechnology
Objectives	➤ To enable students to understand the achievements and principles of Biotechnology, Genetic Engineering, and Recombinant DNA (r-DNA) technology. ➤ To educate students about cloning vectors and their applications in gene cloning technologies.

Detailed Syllabus BTH-75T-551 ANIMAL BIOTECHNOLOGY

UNIT-I

Animal Cell Culture – Basics and Applications: Structure of animal cells, history of cell culture; cell culture media and reagents, culture techniques for mammalian cells, tissues and organs. Types of cultures: primary, secondary, continuous, and suspension. Somatic cell cloning, hybridization, transfection, transformation, and large-scale cell production. Applications in drug testing, toxicity studies, vaccine and pharmaceutical protein production. Animal cell signaling pathway growth factors, 3D organoid culture, and scaffold-based techniques.

15 Lectures

UNIT-II

Cell Lines, Stem Cells, and Medical Biotechnology: Types and preparation of cell lines, stem cells, and stem cell therapy. Veterinary applications in disease diagnosis, vaccine and hormone production. Vaccine development through conventional and recombinant methods, monoclonal antibody production via hybridoma technology, growth hormone production, and phage display technology. Gene therapy principles with examples such as ADA deficiency, Duchenne muscular dystrophy, and cystic fibrosis.

15 Lectures

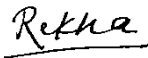
UNIT-III

Transgenic and Genetic Engineering Techniques: Reproductive technologies in animals: artificial insemination, superovulation, in vitro fertilization, and embryo transfer. Gene transfer methods: microinjection, electroporation, lipofection, and viral vectors. Generation and characterization of chimeric, transgenic and knockout animals. Gene editing and silencing using CRISPR-Cas9. Applications in disease models, molecular farming, and studies of genetic disorders.

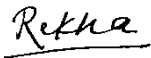
15 Lectures

UNIT-IV

Animal Genomics and Livestock Improvement: Genetic characterization and marker-assisted breeding of livestock. Genome analysis techniques: SNP, STR, QTL. Genetic basis of disease resistance, expression of therapeutic proteins using transgenic animals. Species identification through immunological and DNA-based methods. Detection of adulteration in meat and feed,

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and forensic identification of wild animals using biological samples. Role of CRISPR-Cas9 in animal models and biotechnology in livestock enhancement. **15 Lectures**

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BTH-75P-552 ANIMAL BIOTECHNOLOGY-PRACTICAL

1. Basic Setup of an Animal Tissue Culture Laboratory
2. Preparation of media, sterilization, and practices in a tissue culture laboratory
3. Estimation of RBC, WBC, Hb, and hematocrit value in a given blood sample.
4. Counting of viable cells by TTC methods
5. Isolation of DNA from Blood
6. Trypsinization of monolayer and subculturing
7. Cryopreservation and thawing
8. Measurement of doubling time
9. Role of serum in cell culture
10. Cell fusion with PEG
11. MTT assay for cell viability and growth
12. Preparation of Phosphate Buffered Saline (PBS).
13. Trypan blue dye exclusion test for cell viability.
14. ELISA technique for antigen-antibody detection.

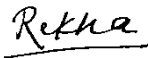
Suggested Books & Text

1. A textbook of Biotechnology-R. C. Dubey, S. Chand & Company Ltd., New Delhi -1996.
2. A textbook on Biotechnology-(n Ed.) H.D. Kumar. EWP - Private Ltd., New Delhi -1998.
3. Animal Biotechnology-M.M. Ranga, Agrobios (India), 2000.
4. Biotechnology-Fundamentals & Applications-S.S. Purohit & S.K. Mathur, Agro Botonics-1999.
5. Biotechnology-V. Kumaresan. Saras Publication-1994
6. Gordon I. 2005. Reproductive Techniques in Farm Animals. CABI
7. Kindt TJ, Goldsby RA & Osbrne BA. 2007. Kuby Immunology. WH Freeman
8. Kun LY. 2006. Microbial Biotechnology. World Scientific
9. Levine MM, Kaper JB, Rappuoli R, Liu MA, Good MF. 2004. New Generation Vaccines. 3rd Ed. Informa Healthcare
10. Lincoln PJ & Thomson J. 1998. Forensic DNA Profiling Protocols. Humana Press.
11. Portner, R. 2007. Animal Cell Biotechnology. Humana Press.
12. Spinger TA. 1985. Hybridoma Technology in Biosciences and Medicine. Plenum Press.
13. Twyman RM. 2003. Advanced Molecular Biology. Bios Scientific.

Course Learning Outcomes:

On successful completion of this course, students will be able to:

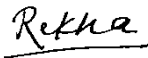
1. Define and explain basic concepts of animal cell culture, types of cultures, and their applications in research and diagnostics.
2. Describe the preparation and use of cell lines and stem cells in veterinary biotechnology and regenerative medicine.
3. Interpret the process of recombinant vaccine development, monoclonal antibody production, and gene therapy with relevant examples.

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4. Explain reproductive and gene transfer technologies like IVF, embryo transfer, electroporation, and microinjection used in animal improvement.
5. Understand the principles and applications of gene editing tools such as CRISPR-Cas9 in creating transgenic and disease-model animals.
6. Analyze the use of molecular markers and DNA-based tools for species identification, meat authentication, and livestock breeding programs.
7. Demonstrate basic biotechnology lab skills, including DNA isolation, cell viability assays, ELISA, and media preparation under sterile conditions.
8. Apply biotechnological knowledge to improve animal health, productivity, and reproduction in the context of veterinary and livestock sciences.

BTH-75T-553 PLANT BIOTECHNOLOGY

Semester	Code of the Course	Title of the Course/Paper			NHEQ F Level	Credits
V	BTH-75T-553	PLANT BIOTECHNOLOGY			7	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Introductory course of Biotechnology				
Course Objectives		➤ To introduce students to the principles and practices of plant biotechnology, including plant tissue culture, genetic transformation, and molecular breeding. ➤ To familiarize students with the applications of plant genomics in crop improvement and biotechnology research.				

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Detailed Syllabus
BTH-75T-553 PLANT BIOTECHNOLOGY

UNIT-I

Fundamentals of Plant Tissue Culture: Historical background, terminology, basic techniques of cell and tissue culture, surface sterilization, and aseptic tissue transfer. Concept of totipotency—dedifferentiation and redifferentiation. Nutritional requirements in vitro, types of nutrient media including inorganic and organic supplements, plant growth regulators, gelling agents, and defined media. Development of explants through somatic embryogenesis and organogenesis. Micropropagation and plant cloning—process, limitations, and applications in agriculture, horticulture, and forestry.

15 Lectures

UNIT-II

Somaclonal Variation and Haploid Technology: Tissue culture variability, somaclonal and other variations—process, advantages, limitations, screening, and verification. Haploid production through androgenesis and gynogenesis, diploidization, and applications. Disease-free plant production using tissue culture—Methods, applications, limitations and virus indexing.

15 Lectures

UNIT-III

Cell and Protoplast Culture: Isolation of cells from plants using mechanical and enzymatic methods, suspension cultures, and single-cell cloning techniques—Bergmann plating, filter paper raft, microculture chamber, microdrop, and feeder layer methods. Protoplast isolation, culture, and fusion—mechanical, enzymatic, and chemical/electrical methods. Protoplast purification, quality assessment, somatic hybridization, and hybridity verification. Use of molecular markers and marker-assisted selection (MAS).

15 Lectures

UNIT-IV

Genetic Engineering and Crop Improvement: *Agrobacterium tumefaciens* and *A. rhizogenes*-mediated transformation, features of Ti and Ri plasmids, DNA transfer mechanisms. Genetic engineering for enhanced photosynthesis, nitrogen fixation, nutrient uptake, and crop productivity. GM crop regulations and biosafety norms in India. Engineering tolerance to biotic stresses (insects, fungi, bacteria, viruses, weeds) and abiotic stresses (drought, salt, temperature, flooding). Quality improvement targeting proteins, lipids, carbohydrates, vitamins, and minerals. Germplasm conservation, identity preservation in GM crops, and bioinformatics tools in plant biotechnology.

15 Lectures

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BTH-75P-554 PLANT BIOTECHNOLOGY -PRACTICAL

1. Preparation of stock solutions of Murashige and Skoog medium
2. Preparation of Murashige and Skoog medium
3. Aseptic handling of explants and Surface sterilization of explants
4. Explant culture
5. Callus culture
6. Anther Culture
7. Demonstration of the Production of synthetic seeds
8. Development of virus-free plants from meristematic tips
9. CTAB method for plant DNA extraction.
10. PCR Demonstration.
11. Agarose gel electrophoresis.
12. *Agrobacterium*-mediated transformation (demonstration).

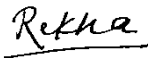
Suggested Books & Text

1. Plant Tissue Culture: Theory and Practice: Theory and Practice by S.S. and M. K. Razdan
2. Introduction to Plant Biotechnology: H.S. Chawla
3. Introduction to Plant Biotechnology – Razdan
4. Plant Biotechnology: The Genetic Manipulation of Plants by Adrian Slater, Nigel W. Scott, Mark R. Fowler
5. Recent advances in Plant Biotechnology and its application- Ashwini Kumar and Sudhir K Sapory

Course Learning Outcomes:

On successful completion of this course, students will be able to:

1. Define and explain basic concepts and terminology of plant tissue culture, including totipotency, organogenesis, and micropropagation.
2. Establish different types of in vitro plant cultures such as callus, anther, and meristem cultures, using standard lab techniques.
3. Prepare nutrient media like Murashige and Skoog medium and calculate media components with accuracy.
4. Compare the environmental impact of genetically modified (GM) plants and understand related biosafety guidelines.
5. Explain the methods of haploid production, somaclonal variation, and their use in developing improved plant varieties.
6. Perform basic techniques such as explant sterilization, synthetic seed production, DNA extraction, PCR, and gel electrophoresis.
7. Understand principles of protoplast culture, somatic hybridization, and the use of molecular markers in plant selection.
8. Apply knowledge of plant genetic engineering (*Agrobacterium*-mediated transformation) for crop improvement and stress tolerance.

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BTH-75T-555 BIORESOURCE WASTE MANAGEMENT

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
V	BTH-75T-555	BIORESOURCE WASTE MANAGEMENT			7	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Introductory course of Biotechnology				
Course Objectives		➤ To examine the various types of solid waste and methods to categorize it. ➤ To find out methods to reduce solid waste at the source. ➤ To understand people's responsibility in reducing and managing waste.				

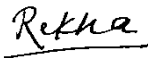
Detailed Syllabus

BTH-75T-555 BIORESOURCE WASTE MANAGEMENT

UNIT-I

Bioresources and Waste Characterization: Bioresources as biomass for food, fuel and chemicals. Biodiversity characterization, circular bioeconomy concept, and GPS applications in bioresource management. Introduction to waste problems, types and categories of solid waste, environmental and health impacts of excess waste generation. Factors influencing waste generation, waste characteristics and analysis. Major issues related to solid waste disposal.

15 Lectures

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UNIT-II

Waste Minimization and Management Techniques: Source reduction strategies, economic benefits of waste minimization, steps to initiate waste reduction programs, and daily operational guidelines. Onsite handling, storage (dustbins, containers, placement), and processing of solid waste. Waste collection methods, equipment, labor needs, and route planning. Processing and disposal through landfilling, composting, incineration, vermicomposting, and bioremediation. Resource recovery and energy generation from waste, reuse in industries, role of fungi and algae in bioremediation and composting of plant and animal waste.

15 Lectures

UNIT-III

Waste Audit and Public Responsibility: Waste terminology, waste analysis methods and waste audit processes. Performance checklists for waste collection, segregation, transport, and treatment. Public responsibility in waste management, Polluter Pays Principle (PPP), Precautionary Principle, Assimilative and Carrying Capacity concepts, Global scenario in scrap trade. Extended Producer Responsibility (EPR), adoption of bioplastics and biodegradable polymers.

15 Lectures

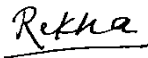
UNIT-IV

Zero Waste and Sustainable Practices: Concept of zero waste living and sustainable lifestyle approaches. Waste reduction strategies at producer, individual, and community levels. Role of awareness, behavioral change, and community participation in promoting sustainable waste reduction. Circular economy and its relevance to zero waste. Role of sustainable consumption, green packaging, and eco-friendly product design. Community-led waste segregation, composting, and recycling models. Importance of policy support, Extended Producer Responsibility (EPR), and Sustainable Development Goals (SDG-12). Emerging innovations in waste-to-resource technologies and upcycling practices.

15 Lectures

BTH-75P-556 BIORESOURCE WASTE MANAGEMENT -PRACTICAL

1. Field report on Biodiversity Park
2. Shelf life management of flowers of importance
3. Study of industrially important plants (Specimens/ products) morphology, botany, and uses
4. Study on the disposal and treatment of biomedical waste
5. Waste water treatment
6. Study on the disposal and treatment of solid waste
7. Study on Waste wastewater treatment plant
8. Visit to Vermiculture unit
9. To study bioresources from plant & animal sources
10. Preparation of compost from organic waste.

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11. BOD and COD measurement from water samples.
12. Isolation of fungi from decaying waste.
13. Bioassay of compost using seed germination.

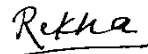
Suggested Books & Text

1. Solid waste Engineering: Principles & management. Issue: G. Techobanogious, H. Theisen, R. Blassen
2. C.L. Mantell, Solid waste management, John Wiley, New York, 1975
3. Bhide and Sudershen, Solid waste management in developing countries
4. Gitanjali Nain Gill, 2011, SAGE Publications' Green Technology: An A-Z Guide (2011), whose work for that encyclopedia formed the basis of her contributions to Britannica. PG Diploma in Waste Management and Environmental Hygiene (PGDWMEH)
5. Hester, R. E. and R. M. Harrison (2002). Environmental and health impact of solid waste management activities. Cambridge: The Royal Society of Chemistry.
6. <https://www.downtoearth.org.in/coverage/costs-and-benefits-of-india-swaste-disposal-options-5623>
7. <https://swachhindia.ndtv.com/national-aluminium-company-limitedadvocates-for-use-of-aluminium-foil-as-alternative-to-plastic-26056/>
8. <https://www.downtoearth.org.in/blog/india-s-challenges-in-wastemanagement-56753>
9. <http://rsos.royalsocietypublishing.org/content/4/3/160764#sec-17>
10. <https://www.downtoearth.org.in/coverage/waste-smart-cities-54119>

Course Learning Outcomes:

On successful completion of this course, students will be able to:

1. Define and classify different types of bioresources and solid waste based on their origin and characteristics.
2. Explain the environmental and health impacts of improper solid waste disposal and the importance of sustainable waste management.
3. Describe various methods of waste minimization, handling, storage, transportation, and disposal including composting, landfilling, and vermiculture.
4. Understand the role of fungi, algae, and microbes in waste bioremediation and compost preparation.
5. Perform basic waste audit techniques and analyze waste samples using standard parameters.
6. Recognize the responsibilities of individuals and industries in waste management through principles like EPR and Polluter Pays.
7. Demonstrate practical knowledge of wastewater treatment, compost preparation, BOD and COD measurement, and fungal isolation from waste.
8. Apply the concept of zero waste and sustainable lifestyle strategies at the community and individual levels to promote environmental responsibility.

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